DISTRIBUTED COMPUTING SYSTEMS COURSE

Phan Trung Phat

Email: phatpt@uit.edu.vn

Department of Computer Networks

University Lecturer at The Faculty of Computer Networks & Communications, UIT - VNU-HCM

Project Instruction

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Outline

1. Introduction to the course
2. Topics
3. Grade/Score Components
4. Reports
5. No. groups/Members per group

1. Introduction

- **Comprehensive Knowledge Base:** Providing foundational and in-depth expertise in Distributed Computing Systems, from core theoretical principles to modern implementation technologies.
- **System Architectures & Models:** Exploring diverse distributed architectures and communication models.
- **Infrastructure & Virtualization:** In-depth coverage of Cluster computing, Cloud computing, Virtualization, and Containerization (Docker/Kubernetes).
- **Data & Service Management:** Leveraging Big Data platforms, Web Services, and Microservices architectures.
- **Emerging Technology Trends:**
  - **Next-Gen Computing:** Serverless, Edge, and Fog computing

    or … Quantum Cloud Computing?
  - **Advanced Integrations:** Distributed AI and Blockchain implementation within distributed environments.

1. Introduction

Week 1: An overview of distributed and parallel computing

Week 2: Cluster Computing

Week 3-4-5: Cloud Computing

Week 6-7: Virtualization and Containerization technologies

Week 8-9: Middleware and Big Data Platforms;

Microservices in distributed systems

Week 10-11: Research and Application Trends

Week 12-13-14-15: Final Project Presentations

2. Topics

- Cloud-native microservices with autoscaling
- Serverless event-driven processing pipeline
- Dynamic Resource Allocation in Cloud Computing / Container-based Platforms
- Load Balancing Techniques for Cloud Resource Management
- Deploy a multi-node cloud service with the open-sources
- Cost & performance comparison: VM vs container vs serverless
- Edge IoT pipeline with cloud offloading
- Hybrid backup and automated failover demo
- Storage benchmarking: object vs block vs file
- Observability & SRE playbook for a cloud service
- Secure cloud application: IAM & zero-trust demo
- Autoscaling in Container / Cloud / Edge Simulation Environments

2. Topics

- You can choose anything you like in Distributed Computing Systems, but you must discuss it with lecturers before registration.

3. Grade

- 25%: Labs
- 25%: Attendance and Project
- 50%: Final Exam

4. Report

- Types
  - Technical Report (Word)
  - Slides (PPT)
  - Demo logs/video/scenarios/results
  - Numerical data
- Structure of Report:
  - Full details of author(s): Full name, StudentID
  - Introduction
  - Description of topics
  - Related work
  - Code analysis/Demonstration
  - References

5. No. groups/Members per Group

- No. groups
  - 14 [total students: 56]
- Members per Group: 4

THANK YOU